Waveguide G502 Augmented Reality Smart Glasses with AI for Efficient Home Interactions

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Abstract

Artificial intelligence (AI) has become the pinnacle of modern technology, allowing humans to operate efficiently and complete tasks more quickly. However, AI has limited daily usage in basic living environments, as AI is usually connected to a mobile phone or home assistant device. The limitation of artificial intelligence usage in living environments reduces the impact AI can have on day-to-day interactions, especially for those who are negatively impacted by disabilities.

Individuals with disabilities also face a daily challenge in home spaces. For example, the blind need assistance with navigation, object recognition, and spatial awareness. The deaf are unable to hear sounds from other people and auditory cues like doorbells, a microwave going off, or a fire alarm. These limitations overall reduce safety, increase the need for outside assistance, and make living extremely difficult.

This engineered solution, *Waveguide G502 Augmented Reality Smart Glasses with AI for Efficient Home Interactions*, will assist those who have disabilities through artificial intelligence, a waveguide AR display, and automated speakers that will enhance the ability for disabled and non-disabled individuals. The glasses use a heads-up display (HUD) that provides an easy way to connect to the AI model. This model takes in information from camera data and environmental audio, which will then convert the data into readable data that can be displayed through the waveguide and speakers. By implementing AI to detect objects, languages, and sounds while using a waveguide AR display module, these smart glasses will be a continuous mobile assistant that can help anyone with basic tasks to life-changing living conditions.

Introduction

The AI industry has been rapidly evolving over the past decade, with modern technologies such as chatbots, facial recognition, and image generation (Brown, 2023). As AI evolves for the future, large companies such as Nvidia and Google have been pushing for a consumer-focused market, where AI models like ChatGPT and Google Gemini become easily accessible to the general public. Recently, at the 2026 Consumer Electronics Show (CES), Even Realities publicly showcased innovative technology using augmented reality through their use of assistance devices such as the Even G2 Smart Glasses, which include real-time conversation and health enhancement.

Assistive technology has also been a growing market for the general public, as technology for home appliances is growing and becoming more popular. There are many assistive devices for disabled individuals, such as canes, guide dogs, and smartphone-based navigation devices for the blind and hearing aids for the deaf. However, these mechanisms lack a critical area of support: real human service. Another example of a lack of reliability within these devices can be seen in the usage and selection of biometric data. This can often be seen in IoT devices, as these devices rely on internet security. This means that if someone is able to access one device, they will gain access to the user’s information. Regarding these issues, this project aims to combat the ongoing problem for handicapped individuals through the use of augmented reality smart glasses that utilize artificial intelligence to support others as if they were their own.

Augmented reality has proven useful in modern daily environments, as companies are able to overlay data to users through technology such as waveguide and birdbath optics (Azuma, 1997). Furthermore, smart glasses, seen heavily in China, have been growing in popularity due to their usefulness in daily tasks and applications. When combined with artificial intelligence, these glasses act like a super tool, often being more helpful than a mobile phone, which is commonly used by almost every person today. Even augmented technology that isn't specifically suited for wearability, such as HUD displays for cars, has crucially helped others with daily tasks.

Previous research has proven that AI-driven appliances have a greater impact on quality of life than algorithm-based models (Reyes, 2020). Additionally, AI models that use datasets such as the Common Crawl dataset are shown to be extremely accurate in the text that is generated from the models. By understanding these past improvements in technology, we can provide a strong foundation to build upon. The implementation of AI models with the Common Crawl in combination with the COCO dataset can be used together in order to design a complex artificial intelligence model algorithm that helps send data to users extremely effectively.

The purpose of this research is to design and evaluate a wearable augmented reality system through the fashionable accessory of glasses with an implementation of AI to support those with disabilities in everyday environments. Specifically, I want to highlight the importance of augmented reality in modern-day society and how it can be a push for a new era of technology and daily usability (Carmigniani, 2011). The Waveguide G502 Augmented Reality Smart Glasses with AI for Efficient Home Interactions specifically aim to target the environment inside a home to improve accessibility, safety, and independence for everyone. By equipping the glasses with multiple functions to help those with disabilities, this engineering prototype will be another foundation for home environments to be completely assisted with AI (Zheng, 2022). By using real-time artificial intelligence models and waveguide displays, these glasses will be a personal assistant that can be taken anywhere without the need for complex motor functions.

Method

Materials

To design the glasses, certain materials were used to create both the frame and the electronics. To design the frame, I used Autodesk Inventor 2026 to outline the model, then printed it using a Bambu Labs P1S 3D printer using basic PLA. PLA is chosen as the primary material due to its lightweight characteristics and rigidity. Additionally, PLA is extremely cheap to manufacture, allowing mass production for consumer needs.

For the internal electronics of the glasses, I used an ESP32-S3 Xiao model for its extremely compact size and processing power. Connected to this PCB are 2 G502 waveguide modules that display augmented reality images into the user's eyes. Included in the glasses are also IMU sensors, speakers, buttons, a scroll wheel, a mini camera, a 24-pin USB-C port, and 3.3V mAh lithium-ion batteries. I tested these electronics with multimeters to ensure the correct voltage was being sent to each device.

To design the AI language model and the software on the augmented reality glasses, I used Android Studio to design an Android app using an SDK. This app uses data from the sensors of the glasses and feeds them through an AI model running on the PCB. Then, the information is fed back into the waveguide modules, which display information to the user. The AI model was trained using the Common Objects in Context (COCO) dataset for image recognition and the Common Crawl dataset for language training. It was coded in a Python environment using Google Colab.

Methods

The design process for the glasses began by designing a frame in CAD. I designed a frame that is compact, lightweight, and fashionable and that also houses the electrical and optical components for the glasses. The glasses frame was CAD-ed in Autodesk Inventor 2026 to ensure the electrical components were properly fitted during rendering. The glasses were then fabricated in the Bambu Labs P1S printer, where, after multiple interactions of the design, the glasses frame was complete and able to store the electronics. I tested the different frames through trial and error, where I tried different thicknesses of the arms to control the functionality and wearability of the glasses.

After I completed the frame, the internal electronics were assembled inside the arms and temples of the frame. In the arms, I put the batteries, PCB, buttons, USB-C port, speakers, microphones, and wiring. The temples of the glasses contained both the waveguide modules and the driver for each module. Finally, the IMU and camera were mounted on the very front of the glasses. I tested voltages in electronics using a multimeter and also tested the accuracy of all the sensors by iterating through multiple readings and comparing them to real-life data, such as volume noise and IMU heading.

The artificial intelligence language model was created using Python, where it was trained using the COCO dataset. It was able to detect home applications such as coffee makers, ovens, sinks, etc. Language processing was developed using the Common Crawl dataset to send text to the user. I tested the accuracy of the data by using a testing portion of the dataset, where the AI model proved to have high accuracy.

Results

Prototype Assembly

The Waveguide G502 AR Smart Glasses were completed after the final iteration of the 3D model. The frame was printed in multiple parts on the Bambu Lab P1S, then assembled using adhesive. The electronics that were mounted inside the arms include the ESP32-S3 Xiao PCB, 3.3V batteries, buttons, scroll wheels, speakers, and microphones. The temples had the G502 waveguide module, the waveguide drivers, and the ribbon cable that connects the PCB to the waveguide. Finally, the nose of the glasses contains the IMU sensor and the camera. The prototype continued to display a bright HUD even in lit environments and stayed operational during testing.

Environment Detection Performance

Environmental noise was detected and observed from the camera and the speakers. Sounds such as doorbells, microwave beeps, and alarms accurately communicated with the waveguide HUD to display output. The algorithms were tested 20 times each at different decibel levels to simulate different noise conditions.

Average Detection Rate: 97.2984%

Most Reliable Sound: Doorbell

Most Unreliable Sound: Microwave beep (usually due to other devices making similar sounds)

System Delay and Response

To ensure that the system is reliable and can be continuously operated for long periods of time, the glasses were tested under several environments. I measured latency between the sensors and the PCB chip using a time algorithm.

Average latency: 121 ms

Average latency over 60 °C: 423 ms

Although the sensor data is much slower at high temperatures, it is still within a small range of error. Furthermore, the glasses take approximately 2.72 hours until the battery is completely drained.

Discussion

The purpose of this study was to evaluate the effectiveness of creating augmented reality smart glasses with AI to create efficient lifestyles for individuals of the general public. The use of this assistive technology demonstrates how integrating artificial intelligence into everyday tasks can allow for the enhancement of both disabled and non-disabled individuals' lives. The completed prototype was able to confirm that a combination of microcontrollers, such as ESP32 Xiao, and the incorporation of processed sensor inputs, including camera data, audio signals, and IMU readings, were able to provide and deliver meaningful information to the waveguide display and speakers. Object recognition through the COCO dataset accurately identified common household items such as sinks, appliances, and furniture in order to demonstrate practical usability in indoor environments. This result shows the greater applications of this solution because it is able to provide real-time spatial and object awareness in the context of providing solutions for the visually impaired. In particular, the speakers and microphones also worked together to detect environmental sounds like alarms or appliances. These sounds could then be converted into visual or audio alerts for the user. This feature is important for users who are deaf or hard of hearing, as it increases safety inside the home. Overall, the results show that the glasses can act as a continuous personal assistant in daily environments. The main goal of this project was to make AR smart glasses that help people in their homes. The results showed that the glasses could recognize objects, hear sounds, and show useful information to the user. This shows that the glasses were able to help with everyday tasks inside the home. Some object recognition errors occurred in low-light conditions or when objects were partially blocked. Processing delays were also observed during complex recognition tasks. Heat buildup inside the glasses frame was another issue due to the compact electronics. To further improve this design, I will remake the frame using a better conductive material and add heat vents to prevent the heat build up inside the glasses. I would also like to experiment with other waveguide modules to test how different modules can help with completing tasks more efficiently.

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Appendix

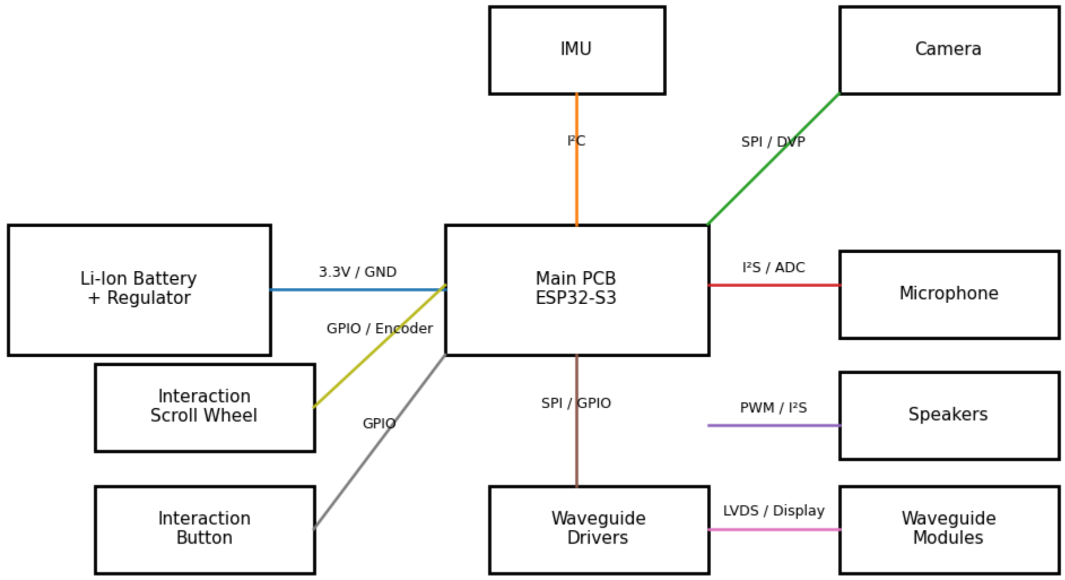
Table 1.1: Accuracy of sound detection in different scenarios

	Dim Room	Bright Room
Fire Alarm	95.025%	97.901%
Doorbell	96.981	98.294%
Microwave Beep	93.052%	97.219

Table 1.2: Latency of sensors under different temperatures

	30 °C	40 °C	50 °C	60 °C
IMU	119 ms	141 ms	203 ms	462 ms
Camera	123 ms	138 ms	214 ms	482 ms
Speaker	120 ms	137 ms	209 ms	459 ms

Figure 1.1: Electronics layout for AR Smart Glasses Design


IMU
Camera
I²C
SPI / DVP
Li-Ion Battery
+ Regulator
3.3V / GND
Main PCB
ESP32-S3
I²S / ADC
Microphone
GPIO / Encoder
Interaction
Scroll Wheel
SPI / GPIO
PWM / I²S
Speakers
GPIO
Interaction
Button
Waveguide
Drivers
LVDS / Display
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Modules